

## §15.8 Triple Integrals in Cylindrical and Spherical Coordinates

### (1) Cylindrical Coordinates (C.C.)

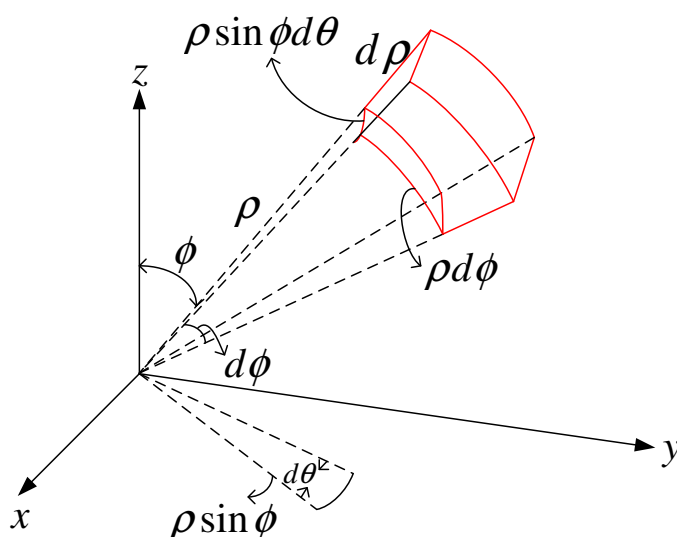
$$E = D \times [u_1(x, y), u_2(x, y)] \quad (\text{圓柱狀適用 C.C.})$$

$D$ : 用 polar coordinate 來表達.

$$\Rightarrow \iiint_E f(x, y, z) dz = \int_{\alpha}^{\beta} \int_{h_2(\theta)}^{h_1(\theta)} \int_{u_1}^{u_2} r f(r \cos \theta, r \sin \theta, z) dz dr d\theta$$

### (2) Spherical Coordinates (S.C.)

$E$  = 球狀物體



Volume element in SC  
 $dV = \rho^2 \sin \phi d\theta d\phi d\rho$

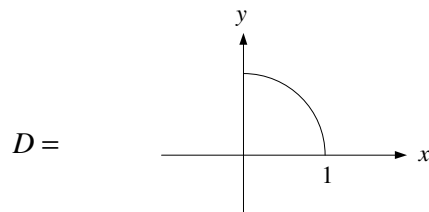
$$\text{Let } E = \{ (\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d \}$$

$$\begin{aligned} & \iiint_E f(x, y, z) dV \\ &= \int_c^d \int_{\alpha}^{\beta} \int_a^b f \rho^2 \sin \phi d\rho d\theta d\phi \end{aligned}$$

**Example 1 :**  $\iiint_{B(\text{單位球})} e^{(x^2+y^2+z^2)^{\frac{3}{2}}} dV$

$$= \int_0^{\pi} \int_0^{2\pi} \int_0^1 e^{\rho^3} \rho^2 \sin \phi d\rho d\theta d\phi = \frac{4\pi}{3} (e - 1)$$

**Example 2 :**  $\int_0^1 \int_0^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} xyz \, dz dx dy$

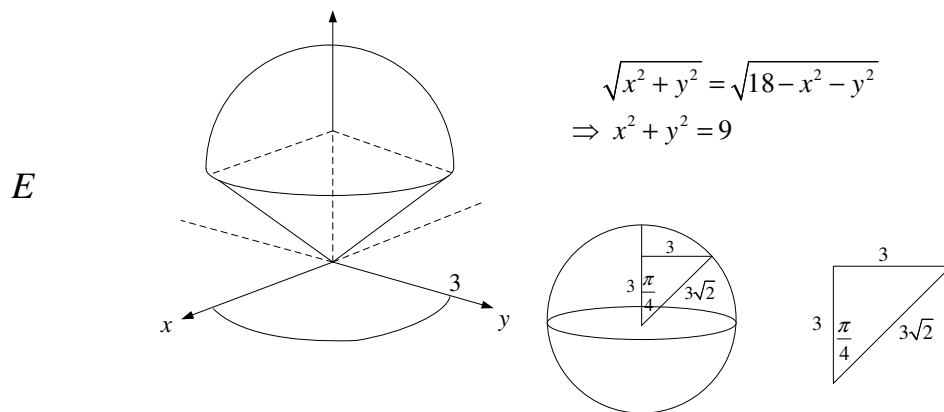


$$= \iint_D \frac{xy}{2} \left[ x^2 + y^2 - (x^2 + y^2)^2 \right] dA$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 \frac{\sin 2\theta}{4} (r^2 - r^4) r^3 dr d\theta = \frac{1}{96}$$

**Example 3 :**  $\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} (x^2 + y^2 + z^2) \, dz dx dy$

**Solution :**

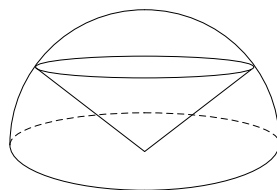


$$= \int_0^{\frac{\pi}{4}} \int_0^{\frac{\pi}{2}} \int_0^3 \rho^2 \rho^2 \sin \phi d\rho d\theta d\phi$$

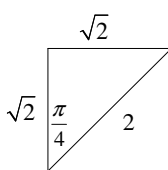
$$= 486\pi \left( \frac{\sqrt{2}-1}{5} \right)$$

**Example 4 :** Find the volume of the solid that lies within the sphere  $x^2 + y^2 + z^2 = 4$ , above the  $xy$ -plane and below the cone

$$z = \sqrt{x^2 + y^2}.$$



**Solution :**

$$\begin{cases} x^2 + y^2 + z^2 = 4 \\ \sqrt{x^2 + y^2} = z \end{cases} \Rightarrow x^2 + y^2 = 2$$


$$\begin{aligned} V &= \int_0^{2\pi} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^2 1 \rho^2 \sin \phi d\rho d\phi d\theta \\ &= \frac{8\sqrt{2}}{3} \pi \end{aligned}$$